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Genome-wide identification and characterization of nucleotide binding site leucine-rich repeat genes in linseed reveal distinct patterns of gene structure

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Abstract: Plants employ different disease-resistance genes to detect pathogens and to induce defense responses. The largest class of these genes encodes proteins with nucleotide binding site (NBS) and leucine-rich repeat (LRR) domains. To identify the putative NBS–LRR encoding genes from linseed, we analyzed the recently published linseed genome sequence and identified 147 NBS–LRR genes. The NBS domain was used for phylogeny construction and these genes were classified into two well-known families, non-TIR (CNL) and TIR related (TNL), and formed eight clades in the neighbor-joining bootstrap tree. Eight different gene structures were observed among these genes. An unusual domain arrangement was observed in the TNL family members, predominantly in the TNL-5 clade members belonging to class D. About 12% of the genes observed were linseed specific. The study indicated that the linseed genes probably have an ancient origin with few progenitor genes. Quantitative expression analysis of five genes showed inducible expression. The in silico expression evidence was obtained for a few of these genes, and the expression was not correlated with the presence of any particular regulatory element or with unusual domain arrangement in those genes. This study will help in understanding the evolution of these genes, the development of disease resistant varieties, and the mechanism of disease resistance in linseed.

Key words: flax, NBS–LRR, in silico gene expression, phylogenetic analysis, promoter analysis, motif analysis.

Résumé : Les plantes font appel à différents gènes de résistance pour détecter les pathogènes et pour induire des réactions de défense. La plus grande classe de ces gènes code pour des protéines dotées de domaines de liaison à des nucléotides (NBS) et de répétitions riches en leucine (LRR). Afin d'identifier des gènes NBS–LRR putatifs chez le lin, les auteurs ont inspecté la séquence génome du lin récemment publiée et ont identifié 147 gènes NBS–LRR. Le domaine NBS a été employé pour produire une phylogénie et ces gènes ont été séparés en deux classes bien connues, soit les gènes apparentés aux TIR (TNL) ou aux non-TIR (CNL), et ont formé huit clades dans un arbre neighbour joining avec rééchantillonnage. Huit structures géniques ont été observées parmi ces gènes. Un arrangement inhabituel des domaines a été observé au sein des membres de la famille TNL, particulièrement au sein du clade TNL-5 de la classe D. Environ 12 % des gènes étaient spécifiques au lin. Cette étude indique que les gènes du lin ont probablement une origine ancienne avec un faible nombre de gènes ancestraux. Une analyse quantitative de l'expression de cinq gènes a montré une expression inductible. Des évidences d'expression in silico ont été obtenues pour quelques-uns de ces gènes et cette expression n'était pas corrélée avec la présence d'un quelconque élément régulateur ou avec un arrangement inhabituel des domaines chez ces gènes. Cette étude aidera à comprendre l'évolution de ces gènes, le développement de variétés résistantes et le mécanisme de résistance aux maladies chez le lin. [Traduit par la Rédaction]

Mots-clés : lin, NBS–LRR, expression génique in silico, analyse phylogénétique, analyse du promoteur, analyse des motifs.

Introduction

Plants have developed a repertoire of resistance (*R*) genes containing various conserved domains. Many such genes conferring resistance to a wide range of pathogens and pests have been identified and cloned from numerous plant species. Based on the presence of conserved domains, the *R* gene products have been grouped into five major classes, viz. detoxifying enzymes, kinases, nucleotide binding site – leucine-rich repeat (NBS–LRR) proteins, extracellular receptors, and receptor kinases (McDowell and Woffenden 2003). Among these, the NBS–LRR class is the most abundant and has been identified from a wide range of plant species, from nonvascular plants to angiosperms (Akita and Valkonen 2002; Kohler et al. 2008; Mun et al. 2009; Xue et al. 2012). The NBS domain of the *R* genes is responsible for the binding and hydrolysis of ATP and GTP (Tameling et al. 2002), whereas LRR is typically involved in protein–protein interactions and is partly responsible for recognition specificity (Leister and Katagiri 2000). In addition, the ARC domain lying between the NBS and LRR domains was identified in potato and reported to play a role in recruitment of the LRR domain to the N-terminal region of NBS to maintain the molecule either in inactive or active states (Rairdan and Moffett 2006).

The plant NBS–LRR genes have further been categorized into two subgroups, TIR and non-TIR, based on the presence (TIR) or absence (non-TIR) of an N-terminal domain with homology to the receptor domain of the innate immunity factors Toll and Interleukin-1 found in animals (Parker et al. 1997). The non-TIR NBS–LRR proteins alternatively contain N-terminal coiled-coil (CC) or leucine zipper (LZ) motifs (Dangl and Jones 2001) and display conspicuous differences in amino acid motifs within the NBS domain. Functions of these domains have not been clearly eluci-

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dated, but they are predicted to be involved in signal transduction pathways (Dangl and Jones 2001; McDowell and Woffenden 2003). The NBS domain of diverse *R* genes is fairly conserved and contains various distinct motifs such as P-loop/kinase-1a, RNBS-A, kinase-2, RNBS-B, RNBS-C, GLPL, RNBS-D-TIR, and RNBS-D-non-TIR (Meyers et al. 1999); of which P-loop, kinase-2, GLPL, and RNBS-D are the most conserved (Xue et al. 2012). These motifs have been extensively utilized to identify resistance gene analogs (RGAs) in model plants and various crop species (Yaish et al. 2004; Palomino et al. 2006) and to understand the genomic architecture of this gene family. Genome sequencing efforts of plant species have facilitated genome-level investigation of the NBS-encoding gene family in various plants, for example, arabidopsis (Meyers et al. 2003; Tan et al. 2007), rice (Monosi et al. 2004), *Medicago truncatula* (Ameline-Torregrosa et al. 2008), poplar (Kohler et al. 2008), grape (Yang et al. 2008*b*), sorghum (Paterson et al. 2009), papaya (Porter et al. 2009), etc. However, very little is known about the NBS–LRR genes in linseed, an oilseed crop from the family Linaceae, as such studies have not yet been reported in this crop.

Linseed or flax (*Linum usitatissimum* L.) is a self-pollinating diploid plant regarded as a cash crop of tomorrow, mainly because of its high nutritional and commercial importance. It is the richest agricultural source of an omega-3 fatty acid, alpha-linonenic acid, and the bioactive compounds, lignans (Dean 2003). Linseed oil is an excellent drying oil used in manufacturing of paints and varnishes, oilcloth, waterproof fabrics, etc. The straw produces fiber of good quality, which is used in making the best quality kerchiefs, material for suites and dresses, bedding, curtains, upholstery, etc. The yield of linseed is severely affected by various fungal diseases caused by *Fusarium* spp., *Alternaria* spp., *Melamspora lini*, etc., and pests such as bud fly (*Dasyneura lini*). Classical genetics studies reported the presence of three resistance genes (*L6*, *M*, and *P*) conferring resistance to *M. lini* in linseed (Dodds et al. 2001). However, this strategy of resistance gene identification is very difficult and slow; as a result, introducing durable disease resistance into agronomically superior varieties has been a major challenge in linseed breeding. Genomic evaluation of disease-resistance gene homologs can help in identifying the putative resistance genes and understanding the mechanism of disease resistance in linseed. Therefore, the main objective of this study was the identification of putative NBS-encoding resistance genes from the linseed genome using bioinformatics approaches, followed by confirmation of expression using quantitative real-time polymerase chain reaction (qRT–PCR). We analyzed the recently sequenced genome of linseed variety CDC Bethune (http://www.linum.ca/) and identified 147 NBS–LRR encoding *R* genes. We examined the predicted linseed NBS–LRR genes for the presence of conserved domains and motifs, and we studied their gene structure, expression, and phylogeny to examine their relationships and evolution.

Materials and methods

Inoculation procedure and sample collection

Seeds of the linseed variety Ayogi (tolerant to *Alternaria* spp.) were surface sterilized using sodium hypochlorite (0.1%) for 15 min, washed thoroughly with sterile distilled water, and sown. The plants were grown in plastic pots containing a mixture of soilless compost in controlled environment chambers at 18 ± 2 °C. The *Alternaria lini* culture was obtained from the Chandra Shekhar Azad University of Agriculture and Technology (CSAUAT), Kanpur, India, and was sent for confirmation to the Fungal Identification Services Division, Agharkar Research Institute (ARI), Pune, India. After confirmation, the isolates were maintained in slants of potato dextrose agar (PDA) at 4 °C. The inoculation procedure and the disease conducive conditions in the growth chamber were as described by Vloutoglou et al. (1999). Linseed plants at growth stage (GS) 5–6 (16–18 true leaves) were artificially inoculated (sprayed until run-off) with the conidial suspension of *A. lini* isolates (10^6 spores/mL). Approximately 20 mL of the conidial suspension was sprayed onto the plants in each pot (10 plants per pot). Mock inoculation was similarly performed using sterile distilled water. The inoculation procedure lasted ~30 min, and at the end of this period no conidial germination was observed. The plants were inoculated at the beginning of a dark period (approximately 1600 hours) and the pots were covered with plastic bags to maintain high humidity and then wrapped with aluminum foil to create darkness. Samples from inoculated and mock-inoculated plants were collected at 0, 4, 7, and 10 days after inoculation (DAI), immediately frozen into liquid nitrogen, and stored at −80 °C until use.

Identification of NBS–LRR genes in linseed

We used all the 47 912 predicted gene models of the linseed genome (http://www.linum.ca/) to identity the NBS–LRR genes. The gene models were searched against the Pfam database using Pfam ver. 24 software with an *e*-value of 0.1 (Bateman et al. 2002), and the results were analyzed to identify TIR-, NBS-, and LRR-containing sequences. The NBS-containing sequences were further analyzed using COILS software (Lupas et al. 1991) at a threshold of 0.9 to detect CC domains. Alternatively, the 114 curetted *R* gene sequences (http://www.prgdb.org/) were also searched against the 47 912 predicted gene models of linseed with blastp, using a threshold *e*-value of 10^{-15}. The obtained hits were searched against the Pfam database and COILS software as described above to identify TIR, NBS, LRR, and CC domains. Nonredundant NBS–LRR genes were determined by eliminating the duplicates.

Identification of homologs of linseed NBS–LRR genes

The whole genome protein sequences of arabidopsis and poplar were downloaded from ftp://ftp.tigr.org/pub/data/a_thaliana/ath1/SEQUENCES and http://genome.jgi-psf.org/Poptr1/Poptr1.download.ftp.html, respectively, and protein databases were created on an in-house server. Viroblast (ver. 2.2+) (Deng et al. 2007) was configured to use the stand-alone suite of BLAST programs (ver. 2.2.25+) and was used to search the arabidopsis and poplar protein databases with the predicted protein sequences of linseed *R* genes, as well as the predicted NBS part of these genes, using blastp. The sequences of arabidopsis and poplar orthologs with the highest match scores were imported in Microsoft Excel (ver. 2007) and further analyses were performed.

Phylogenetic analysis of linseed NBS–LRR genes

Only the NBS domain was used for phylogenetic analysis, as it was conserved in both CNL and TNL genes. The NBS domain sequences of all the genes were extracted and the sequences having NBS domain length of ≥150 aa were used for alignment. HMMER3 software (http://hmmer.janelia.org) was initially used for building the profile of NBS domain and then for alignment. The alignment was edited using MEGA ver. 5.0 software (Tamura et al. 2011), and the sequences containing fewer than 75% of the hidden Markov model (HMM) match-state residues were retained for subsequent analysis. Indels and poorly aligned regions were removed by trimming the regions outside the HMM match-states prior to phylogenetic tree construction. Phylogenies were calculated using parsimony and bootstrapped neighbor-joining algorithms. The trees were constructed using protpars in the Phylip suite ver. 3.6 with a bootstrap value of 100 using the Mobyle portal (http://mobyle.pasteur.fr/). The input sequence order was jumbled five times and topologies were calculated based on each sequence order. One most-parsimonious tree was chosen at random to serve as the basis for branch length calculations. Maximum likelihood branch lengths were calculated on the parsimony topologies with TreePuzzle ver. 5.2 (Schmidt et al. 2002) using the substitution model of Müller and Vingron (2000). Amino acid frequencies were calculated from the input trees, and rate heterogeneity was allowed with four γ rate categories. A neighbor-joining tree of the cleaned alignment from hmmalign was calculated using MEGA ver. 5.0 (Tamura et al. 2011) with 1000 bootstrap replicates. In addition, two more trees were constructed using (*i*) arabidopsis, poplar, and linseed NBS–LRR sequences, and (*ii*) curetted NBS containing *R* gene sequences (http://www.prgdb.org) and the linseed NBS–LRR sequences.

Analysis of conserved motif structures

The domain and motif analyses were carried out to explore the structural diversity among the predicted NBS–LRR genes. Accordingly, the sequences were divided into three classes as those containing the N-terminal motif (sequence upstream the P-loop of NBS domain), the NBS domain (P-loop to WMA), and the LRR-C-terminal domain (sequence downstream the WMA of NBS domain). The three classes of sequences were separately analyzed using the MEME/MAST system (http://meme.sdsc.edu/meme/website/intro.html) (Bailey et al. 2006) to investigate the protein motifs in more detail. MEME (multiple expectation maximization for motif elicitation) motif analyses were performed on each of the subgroups separately (e.g., TNL (containing the domains TIR, NBS, and LRR), CNL (containing the domains CC, NBS, and LRR)) and with settings designed to identify 15, 20, 30, or 50 motifs. MAST (motif alignment and search tool) (Bailey and Gribskov 1998) was used to assess the correlations between MEME motifs and the associated distance matrices.

Analysis of promoter regions of NBS–LRR genes

For each NBS–LRR putative gene, a 2 kb upstream region was selected according to the position of the genes provided by Gbrowse annotation (http://www.linum.ca/) on the scaffold sequences of linseed. The extracted sequences were screened against the PLACE database (http://www.dna.affrc.go.jp/PLACE/) to identify the *cis*-acting regulatory elements. The regulatory elements overrepresented in the dataset and those known to be involved in regulation during the resistance response and under stressed conditions were selected for further analysis (Jang et al. 2006). Among them, WBOX (TGAC(C/T)), CBF (GTCGAC), DRE ((G/A)CCGAC), and GCC boxes were retained for further analysis, as they were reported to be present in the promoters of resistance genes (Jang et al. 2006).

In silico and qRT–PCR expression analysis

The putative NBS–LRR genes were BLAST searched against the NCBI linseed EST dataset (accessed 11 April 2011; 286 894 sequences, http://www.ncbi.nlm.nih.gov/nucest?term = Linum%20usitatissimum) to obtain transcriptional evidence for individual NBS–LRR genes and to estimate the number of ESTs expressed per tissue type and gene model. We used >85% sequence identity criterion for any EST to map onto a gene model. These tissue types include flower (FL), seed coat at globular stage (GC), pooled endosperm (EN), seed coat at torpedo stage (TC), heart embryo (HE), globular embryo (GE), torpedo embryo (TE), bent embryo (BE), mature embryo (ME), etiolated seedling (ES), stem (ST), leaf (LE), peeled stem (PS) (Venglat et al. 2011), 12 DAF bolls, and outer fibrous stem tissue.

Total RNA from both inoculated and control samples was extracted using Spectrum Plant Total RNA kit (Sigma-Aldrich, USA) and treated with DNaseI (Promega, USA). The RNA was reverse transcribed using MultiScribe reverse transcriptase (Applied Biosystems, USA) and oligo(dT) primer. Nineteen gene models were randomly selected for quantitative expression analysis. Primers spanning various regions of RGAs were designed using Primer3 (Rozen and Skaletsky 2000). qRT–PCR was carried out in a 7900HT Fast real-time PCR system (Applied Biosystems, USA). Each 10 μL qRT–PCR cocktail contained 0.25 μmol/L each of forward and reverse gene-specific primers, 4 μL of 1:10 diluted first strand cDNA, 1× FastStart universal SYBR green master mix (Roche, USA), and sterile milliQ water. The cycling conditions used were as follows: 95 °C denaturation for 10 min, followed by 40 cycles of 95 °C for 3 s, primer annealing at 55 °C for 15 s, and extension at 60 °C for 30 s. Following amplification, a melting dissociation curve was generated using a 62–95 °C ramp with 0.4 °C increment per cycle to monitor the specificity of each primer pair. The eukaryotic translation initiation factor 5A (*ETIF5A*) gene from linseed was used as a reference or housekeeping gene for the qRT–PCR (Huis et al. 2010). The housekeeping gene was selected after confirming its stability of expression across the tissues used in the study. Two technical replicates for each of the three biological replicates were performed. PCR efficiencies of the housekeeping gene and the gene of interest were calculated using LinRegPCR (Ramakers et al. 2003), and the qRT–PCR conditions were optimized such that the efficiencies were similar and close to 2.0. Relative transcript abundance calculations were performed using comparative CT ($^{\Delta}$CT) method described by Schmittgen and Livak (2008).

Results

Mining the linseed genome for NBS–LRR genes

The scaffold assembly of the linseed genome (http://www.linum.ca/) contained 302 Mb of nonredundant sequences, representing an estimated 85% genome coverage. This coverage is consistent with the length of the entire low-copy fraction previously estimated by reassociation kinetics and captures nearly the entire gene-rich portion of the linseed genome (http://www.linum.ca/). The 47 912 linseed gene models were searched against the Pfam database to identify TIR, NBS, and LRR domains. Additionally, the gene models were also searched using the 114 curetted *R* gene sequences from PRGdb (http://www.prgdb.org/) as blast queries using an *e*-value of 10^{-15}, which identified 2220 *R* gene like sequences. These hits were searched against the Pfam database and COILS software to identify TIR, NBS, LRR, and CC domains. The results of both the analyses were the same, and 147 nonredundant NBS–LRR encoding genes were identified from the linseed genome (supplemental Table S1).[1] To identify partial and truncated NBS–LRR genes, we considered a minimum size of 24 amino acids for the NBS domain.

Gene structure and intron–exon configurations of linseed NBS–LRR genes

We analyzed the gene structure, intron positions, and phases of the linseed CNL and TNL genes to assess the diversity within and between each family (Table S4). The intron–exon boundaries of the linseed genes (available at http://www.linum.ca/) were analyzed for the 147 NBS–LRR genes. To determine their intron–exon configurations, a blastp search was carried out using the full protein sequence, as well as only the NBS region, of the predicted linseed R proteins against all the arabidopsis proteins. The structures of the top matched arabidopsis genes were compared with those of the corresponding linseed genes, which revealed that the number of introns was less in CNLs (0–13) than in TNLs (2–16) (Tables S4 and S5; Fig. 1).

Distinct patterns of gene structure were identified in CNLs (CNL-A (8) and CNL-C (40)) and TNLs (TNL-B (21), TNL-C (13), and TNL-D (56)) (Table 1 and Fig. 1). Most CNLs (25) showed a common gene structure as observed in the arabidopsis CNL-C protein At3g14470. The TNL family genes showed complex structures usually consisting of some short exons in the LRR region and most of them were interrupted by a phase-0 intron (Fig. 1). In these, various domains (TIR, NB-ARC, and LRR) were often separated by introns. Interestingly, the linseed proteins, which matched with

[1]Supplementary data are available with the article through the journal Web site at http://nrcresearchpress.com/doi/suppl/10.1139/gen-2012-0135.

Fig. 1. Comparison of the intron–exon configuration of selected *R* genes from linseed (L) with related arabidopsis (A) genes. Numbers above exons indicate the size of the exons in base-pairs. Numbers between exons denote intron phases.

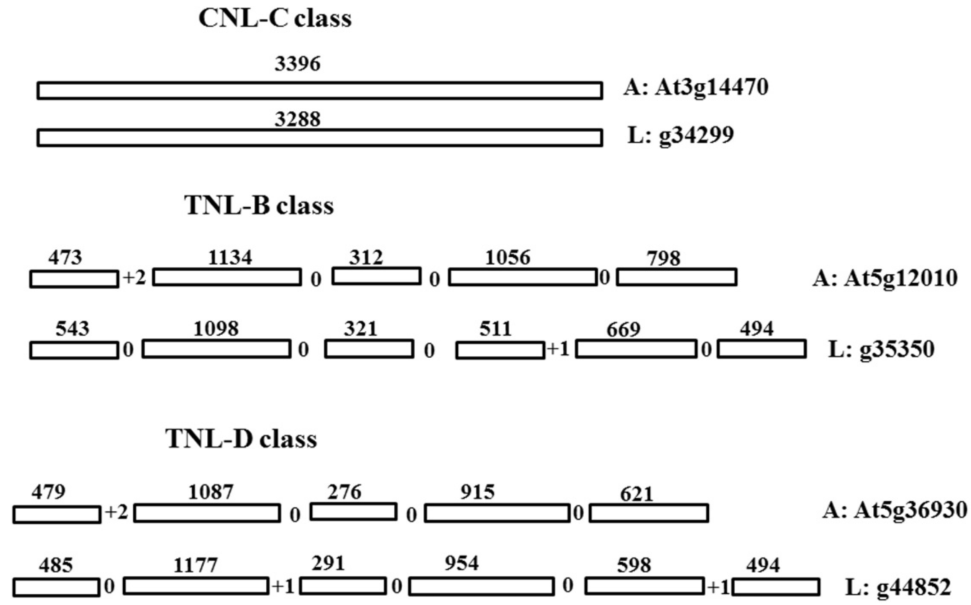


Table 1. NBS–LRR genes classified into different classes and their domain compositions.

	Domains																
Class	CN	CNL	CNNL	CNNNL	N	NL	TCNL	TLTNL	TN	TNL	TNLTNL	TNN	TNNL	TNNN	TNTNL	TTNL	Total
CNL																	
CNL-A	2	5	—	—	1	—	—	—	—	—	—	—	—	—	—	—	8
CNL-B	—	—	—	—	1	—	—	—	—	—	—	—	—	—	—	—	1
CNL-C	8	26	2	1	1	2	—	—	—	—	—	—	—	—	—	—	40
TNL																	
TNL-A	—	—	—	—	1	—	—	—	—	—	—	—	—	—	—	—	1
TNL-B	—	—	—	—	1	—	—	—	6	12	—	1	—	—	1	—	21
TNL-C	—	—	—	—	1	2	—	—	2	8	—	—	—	—	—	—	13
TNL-D	—	—	—	—	1	1	2	1	8	36	1	2	1	1	1	1	56
TNL-H	—	—	—	—	—	—	—	—	6	1	—	—	—	—	—	—	7
Total	10	31	2	1	7	5	2	1	22	57	1	3	1	1	2	1	147

high confidence to a type A protein (At4g36140) from arabidopsis having unusual structure (TNTNL), had only TN or TNL domains. Similarly, the linseed *R* genes (*g43650*, *g43708*, and *g8052*) having high similarity to the At1g27170 arabidopsis protein (a TNL-C protein with canonical structure) either lacked the LRR domain or had unusual structures.

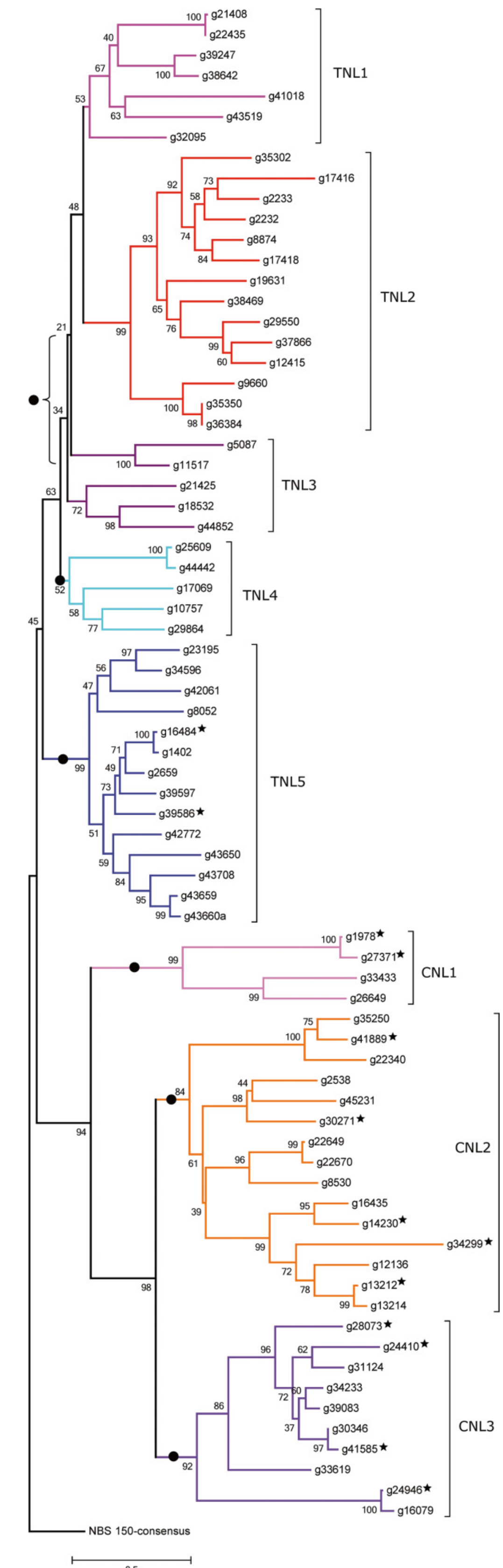
Phylogenetic analysis of NBS–LRR genes

To assess the sequence diversity and evolution of the linseed *R* genes, a phylogenetic tree was constructed using only the NBS domain of linseed R proteins, as it is conserved in both CNL and TNL proteins and contains numerous conserved motifs (Fig. 2). The NBS consensus sequence developed from the extended HMM model was used as an out-group to root the tree. To understand the evolution of linseed genes, arabidopsis (44) and poplar (104) proteins having high sequence similarity to the linseed TNL and CNL proteins, as well as the curetted NBS containing R proteins from PRGdb (http://www.prgdb.org), were included in the phylogenetic analysis. Both parsimony- and distance-matrix-methods yielded very similar results (data not shown). From the values obtained in the bootstrap analysis, it was apparent that most of the deep nodes of the tree have high statistical significance. The tree showed long branch lengths and closely clustered nodes, reflecting a high level of sequence divergence (Fig. 2).

As reported in earlier studies, the phylogenetic tree was divided into two main clusters: TNL and CNL (Meyers et al. 1999), which were further divided into five TNL (TNL-1, TNL-2, TNL-3, TNL-4, and TNL-5) and three CNL (CNL-1, CNL-2, and CNL-3) clades (Fig. 2). Interestingly, the truncated *R* genes (TN, CN, NL, and N) were dispersed among various clades throughout the tree, indicating a diverse rather than a monophyletic origin due to domain loss. In the CNL family, CNL-2 was the largest clade (15) and was composed of the type C linseed R proteins (Fig. 2). It mainly contained poplar R proteins and five known R proteins of diverse origin (Fig. S1). The CNL-3 clade was also composed of type C linseed R proteins (10) and showed high similarity with arabidopsis RPM1 R protein. Interestingly, all the linseed R proteins of CNL-1 clade (4) were of type A and showed high similarity to arabidopsis R proteins (At5g66920 and At4g433300). However, they did not cluster with any characterized R proteins (Fig. S2).

Within the TNL family, clades TNL-2 (14) and TNL-5 (14) contained the majority of the R proteins. The TNL-2 clade composed of mainly B-type R proteins clustered with the well characterized P2 protein of linseed and poplar R protein; whereas, TNL-5 clade composed of both C- and D- (three with unusual domain architecture) type R proteins, in equal proportion, showed high similarity to the previously characterized linseed *R* genes (L6: AAA91022 and M: AAB47618). The clades TNL-1 and TNL-3 clustered with WRKY

Fig. 2. Neighbor joining bootstrap tree of NBS domains of NBS–LRR predicted proteins from linseed. The dots indicate approximate coalescence points for NBS sequences from arabidopsis and (or) poplar. The * indicate truncated genes.



transcription factor 52 of arabidopsis and resistant proteins from the family Solanaceae, respectively. Both clades composed of TNL-D-type linseed R proteins clustered mainly with poplar R proteins, except one linseed R protein with unusual domain architecture (g21425; TNLL), which clustered with arabidopsis R proteins. The TNL-4 clade contained five R proteins and appeared to be paraphyletic in origin (Fig. 2) and did not cluster with any characterized R proteins. Interestingly, the D-type R proteins were observed in all the TNL clades.

Conserved domains and motifs in linseed NBS–LRR genes

Various domains used for classification of NBS–LRR genes such as TIR, NBS, LRR, CC, etc. were identified in the putative 147 NBS–LRR genes as described before. Based on the Pfam analysis of the predicted proteins, as well as homology with previously described motifs within the NBS domain (Meyers et al. 1999), the 147 genes were divided into two major families: CNL (49) and TNL (98) (Table 1). From the 49 CNLs, 13 sequences lacked LRR (CN and N), while five sequences were devoid of CC domains (NL and N). The fusion of RpW8 and PRK domain with the C-terminal region of genes belonging to CNL family was observed (Table S1). On the contrary, diverse domain arrangement was observed in the TNL family. The most common type was typical TNL (57/98, 58.17%) followed by TN (22/98, 22%) (Table 1). In addition, other domains, viz. F box, gal-binding lectin, and pg-box, were also identified in the gene models of TNL class.

Further, the N-terminal, NBS, and C-terminal regions of the putative linseed *R* gene proteins were analyzed separately using the program MEME (Bailey et al. 2006). The N-terminal region ranged from 148 to 248 and from 144 to 1168 amino acids in the CNL and TNL families, respectively; both TIR and CC domains were identified in this region. Fifteen distinct motifs were observed in the N-terminal and NBS regions of CNL and TNL families (Tables S2 and S3). Several conserved motifs (TIR 1–4) were observed and their order was also conserved in the N-terminal region of the TNL linseed R proteins (Table S2). The length of the NBS domain ranged from 243 to 294 amino acids and the crucial motifs of the NBS domain (P-loop/kinase-1a, kinase-2, kinase-3a, and GLPL) were highly conserved among the predicted linseed R proteins. On the contrary, the LRR C-terminal region was highly variable in sequence and size (1–898 amino acids in CNL and 26–1854 amino acids in the TNL family), and 50 motifs were identified in both CNL and TNL sequences.

Promoter region analysis of NBS–LRR genes

To understand regulation of the NBS–LRR genes, a 2-kb region upstream of the predicted NBS–LRR genes was used to identify promoter sequences (Table S7). Four regulatory elements (WBOX cassettes, Type II MYB and G boxes associated with the WRKY transcription factors (Dong et al. 2003); CBF and DRE boxes (Sakuma et al. 2006); and the GCC motif associated with the ERF-type transcription factors (Ohme-Takagi et al. 2000)) were overrepresented. These regulatory sequences are associated with responses to biotic or abiotic stresses. Among all, WBOX elements were the most abundant, averaging 4.59 for the CNL and 4.49 for the TNL families. Of the predicted NBS–LRR genes, 66% contained between 4 and 10 copies of WBOX elements, with four and six copies being the most common. On the contrary, the average numbers of other element types were 0.62 (CBF), 0.62 (GCC), 0.28 (DRE), and 1.24 (associated with WRKY transcription factors) and observed only once per upstream region. In a few cases, multiple boxes of these regulatory elements were observed. Interestingly, the WBOX elements were fairly uniformly distributed across the tree (Table S7). No groups showed systematic excesses or deficits of the less numerous CBF, GCC, or DRE box motifs. No significant correlation between the arrangement of these promoter sequences (WBOX, CBF, GCC, and DRE) and in silico expression was observed (Table S7).

Table 2. Primers used for qRT–PCR analysis.

Gene model	Forward (5′–3′)	Reverse (5′–3′)
g13214	TGTCTGCCGGAGTGGCTTCA	TCCTTGAGTGTGTGCATCCA GATTG
g35350	ACCTCAGCTTCGTCGGTTGC	CCCACCGTACGGAGATTGGA
g16484	TGTCCGTTGTGAAGGGTGGA	CCTCCGAGCGATTGTCATCA
g36384	CGGATTCGTCACAGCCCTCT	GGGACAACGGTCAGGTTTGC
g2659	GAAGCATCCTCGTCGTGTGC	CGCAAACAGCGTCAGTCCAT

In silico and qRT–PCR expression analysis

In silico expression analysis of the identified NBS–LRR genes was performed by comparing these genes against the 286 894 EST sequences from 13 libraries downloaded from GenBank (11 April 2011). A criterion of >85% nucleotide identity between ESTs and NBS–LRR genes was applied to gain EST support for the genes. In cases where one EST matched with multiple candidate genes, only the gene with the highest alignment score was considered. At this threshold, 35/147 genes (representing 23% of the predicted NBS–LRR genes) exhibited EST support (Table S6). In all, 56 EST matches were identified, and the frequency of ESTs varied from 1 to 10 per NBS–LRR gene model. These genes were expressed in a wide range of cDNA libraries, including those constructed from various development stages and tissue types. Among the different tissue types, peeled stem (PS; 32%) had the largest number of expressed genes. The percentage of the genes expressed per clade varied from 2.7% to 47% and even between highly similar genes within the same group (Fig. 2; Table S6). A majority of the TNL genes had EST support, and the highest of 18 ESTs were mapped to seven genes of the TNL-5 clade, whereas only three ESTs mapped to three genes from the CNL-2 clade. Among all the expressed genes, *g2659* showed highest expression in stem (ST), leaf (LE), and mature embryo (ME) (Table S6).

The qRT–PCR was performed to investigate expression of the putative NBS–LRR genes. Five of the 19 gene models (Table 2) showing stable and consistent amplification were selected for quantitative expression analysis. The expression profiles of the selected gene models are presented in Fig. 3. The highest level of expression was observed at 4 and 7 DAI in inoculated tissues, while in the control tissue, *g2659* over-expressed at 7 and 10 DAI, although its expression was higher in other control tissues as well.

Discussion

Linseed has been grown since ancient times for fiber (flax) and oil-rich seed (linseed). Such a long cultivation history of linseed in a wide range of environmental conditions and exposure to various biotic and abiotic stresses suggests that numerous *R* genes for diverse biotic stresses might be present in the linseed genome. Therefore, there is a need for a better understanding of the genomic organization of NBS–LRR encoding genes in the linseed genome and characterization of their genetic and molecular mechanisms. Recently, abundant genomic resources in the form of ESTs and genome sequence have been developed for this crop (Ragupathy et al. 2011; Venglat et al. 2011). We used these resources for genome-wide discovery of NBS-encoding *R* genes from the linseed genome. Analysis of predicted domain structure, promoter regions, phylogeny, and in silico gene expression revealed a number of prominent features as reported for NBS–LRR genes from the other plant species.

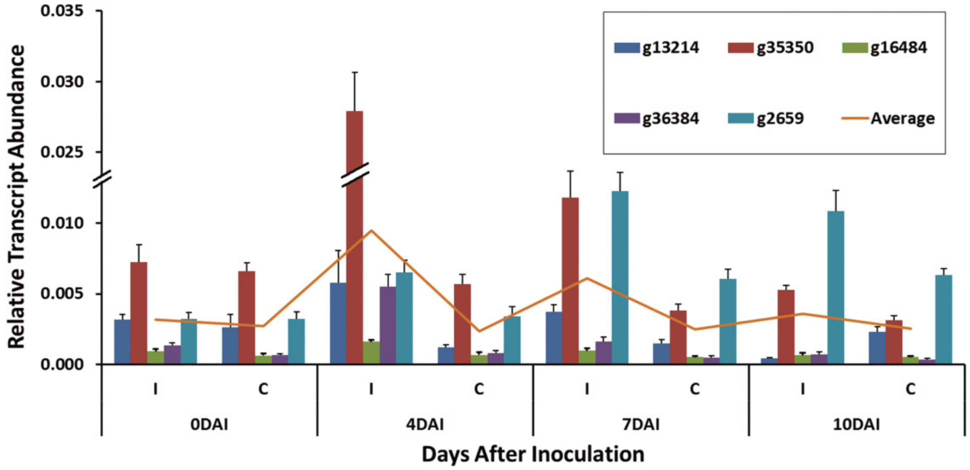
Evolution of linseed NBS–LRR genes

We identified a total of 147 NBS–LRR coding genes, representing about 0.3% of all the predicted linseed proteins. This percentage is less than that reported in other plant genomes (0.6%–1.8%) (Meyers et al. 2003; Monosi et al. 2004; Yu et al. 2005; Ameline-Torregrosa et al. 2008; Kohler et al. 2008; Yang et al. 2008*b*). This could be due to several reasons. (*i*) Assembly of short sequence reads—we used the genome sequences assembled from Illumina short sequence reads (44–75 bp) with 95× coverage (http://www.linum.ca). However, the NBS–LRR genes are reported to be multicopy and highly diversified genes (Kuang et al. 2004; Yang et al. 2008*a*) and cannot be assembled yet by short sequence reads. Hence, the linseed genome might contain a larger number of NBS–LRR genes than the 147 such genes identified in this study. (*ii*) Misannotation—nearly 50% of the initially identified genes (73 out of 147) were smaller in size and possibly resulted because of missed start and (or) stop codons. About 36% of errors in automated annotations were also observed in arabidopsis (Meyers et al. 2003). (*iii*) Deletion of redundant NBS-encoding gene components after ancient whole genome duplication, as reported in other plants (rice (Yu et al. 2005), grape (Yang et al. 2008*b*), arabidopsis (Nobuta et al. 2005), and poplar (Kohler et al. 2008)). Likewise, it has been suggested that linseed might have undergone an ancient polyploidization event (Ragupathy et al. 2011). (*iv*) Distribution of NBS–LRR genes—it is well known that most NBS–LRR coding genes are unevenly distributed and clustered in specific chromosomal regions (Meyers et al. 2003; Monosi et al. 2004; Ameline-Torregrosa et al. 2008). (*v*) Linseed may harbor a smaller number of NBS–LRR genes compared to other plant species. As the linseed genome is yet to be fully sequenced, there is a possibility that the genomic regions containing major NBS-encoding gene clusters are yet to be sequenced. This percentage might change when the linseed genome is fully sequenced and annotated. However, the estimates are similar to those obtained in *Brassica rapa* (Mun et al. 2009) as well as papaya (Porter et al. 2009).

As observed in other dicots, phylogenetic analysis of linseed NBS–LRR genes showed two distinct families, the TIR–NBS–LRR related genes and the non-TIR NBS–LRR related genes, reflecting the well-known ancient differentiation of NBS–LRR genes into two major families (Meyers et al. 2003). Each family was further divided into multiple clades based on their clustering in phylogenetic tree and classes based on distinct domain organizations as described by Meyers et al. (2003) in arabidopsis (TNL-A–TNL-D and CNL-A and CNL-C). The phylogenetic tree indicated that the linseed NBS–LRR genes probably originated from a small number of progenitor sequences in contrast to that in arabidopsis and poplar. Alternatively, it is also possible that many progenitor genes might have subsequently been lost during linseed evolution. The above clades are linseed specific as they contain very few of the arabidopsis or poplar *R* genes. However, this needs to be validated by including more *R* genes from other closely related families of the order Malphygials. We could not attempt this ourselves because of the lack of genomic sequence data of these family members.

Many of the linseed *R* genes showed high sequence similarity in some clades (e.g., TNL-2 and CNL-3) indicated by the presence of many sequences with small branch lengths. The CNL family of proteins appeared to be more ancient than the TNL family proteins owing to their long branch lengths. Similar results have also been reported for other plant genomes (Meyers et al. 2003; Kohler et al. 2008). Within seven CNL family genes, C-terminal fusion of Rpw8 domain was observed. Similarly, C-terminal fusion of Rpw8 domain has also been observed in the *CU013515_1.4/Mt5g1164* gene of *M. truncatula* (Ameline-Torregrosa et al. 2008) and *At5g66910* gene of arabidopsis. Thus, it appears that these genes have remained as single copy genes in their respective genomes since the last ancestor of these plant lineages. The *Rpw8* gene in arabidopsis and other dicots provides broad-spectrum mildew resistance (Xiao et al. 2001). Thus, it can be presumed that such genes in linseed might also provide a broad spectrum disease resistance. In general, the CNL proteins are predicted to have basic defense roles as regional adaptation to the specific biotic stresses (Yang et al. 2008*b*).

Fig. 3. Expression profiles of five *R* genes in inoculated and control linseed tissues collected at 0, 4, 7, and 10 days after inoculation (DAI).



The TNL genes in linseed share the largest proportion of the NBS–LRR genes and show large domain diversity compared with the CNL genes, as also reported in arabidopsis (Meyers et al. 2003), *M. truncatula* (Ameline-Torregrosa et al. 2008), and poplar (Kohler et al. 2008). Such greater diversity can be explained partly by the intron–exon and domain structure of TNL family genes. The linseed TNL proteins contained more introns than CNL proteins as observed in arabidopsis (Meyers et al. 2003), poplar (Kohler et al. 2008), and cereals (Bai et al. 2002). Most of the additional introns and domains in these TNL genes occur at the 3′ end of the genes. It has been proposed that each of the TIR, CC, NBS, and LRR domains evolved independently with distinct functions and later fused to form new proteins. For example, the potato CNL protein Rx can act in *trans* with the CC or the LRR domains expressed from separate proteins to produce a hypersensitive response (Moffett et al. 2002).

Interestingly, most of the unusual domains identified in the present study were observed in the TNL-5 clade and all were type D proteins. Thus, this clade or the type D proteins seem to tolerate more domain arrangements than other groups. The extra domains present in the TNL-D might have fused during the gene/domain duplication or recombination events resulting in the diversity of the proteins. The evidence of exon additions or fusions was observed in WRKY-related domains and some metallopeptidases in arabidopsis (Meyers et al. 2003) as well as BED/DUF1544 domain in poplar (Tuskan et al. 2006) genomes. The presence of TNTNL in arabidopsis is predicted to arise from the fusion of TN and TNL genes. Interestingly, linseed TNL-A proteins showing high similarity to the arabidopsis TNTNL protein (At4g36140) exhibited either TN or TNL domains, further supporting the evidence of exon fusion hypothesis. The occurrence of diverse genes within a single clade in the TNL family (e.g., TNL-5 contains both type C and D proteins) was also observed. This might have resulted from clustered arrangement of NBS–LRR genes and their recombination. Similar clustering of diverse genes has also been observed in arabidopsis (Meyers et al. 2003). Overall, the high diversity of TNL family genes indicates its role in recognizing species-specific pathogens (Yang et al. 2008*b*).

Several classes of truncated *R* genes (CC–NBS (CN), TIR–NBS (TN), and NBS–LRR/NBS (NL/N)) were also observed in linseed. These truncated proteins were dispersed among various clades and classes, and they were scattered throughout the phylogenetic tree (Fig. 2). It is possible that some of these genes were misassembled or some additional unique domains are yet to be identified. Such truncated *R* genes are suggested to act as adaptor molecules in the resistance response through recruitment or interaction with NBS–LRR genes (Belkhadir et al. 2004). Similarly, NBS proteins lacking an LRR were also identified in arabidopsis and *M. truncatula* (Meyers et al. 2003; Ameline-Torregrosa et al. 2008). Genes containing only the NBS domains were also observed, which are reported to be either absent or reduced in other dicot genomes (Bai et al. 2000). However, three such truncated genes (*g1978*, *g16484*, and *g27371*), interestingly, showed expression evidence based on the EST data of linseed (Table S6). Furthermore, the unusual TIR–NBS-containing genes (TNCL and TCN), similar to those previously reported in arabidopsis (Meyers et al. 2003) and *M. truncatula* (Ameline-Torregrosa et al. 2008), were also observed in linseed. In addition, few other domains were also identified, indicating that the gain of these domains has occurred in this family of proteins in the linseed lineage (Table S1).

Origin of the linseed NBS–LRR genes

The coalescent points on the phylogenetic tree provide relative time references (Ameline-Torregrosa et al. 2008), indicating the clades that have probably expanded in linseed. The linseed R proteins were clustered with similar proteins from poplar, a closely related species, suggesting that they evolved from a common ancestor and later expanded separately within each species (Fig. S1). The TNL family proteins show recent expansion compared with the CNL family proteins as evident from the tree. As indicated by the coalescent points, almost all the clades from TNL family expanded at similar times, whereas variation in the time scale was observed in the expansion of CNL family genes. When phylogenetic analysis of linseed R proteins was carried out with the curetted R proteins having NBS domains (http://www.prgdb.org), various clade proteins clustered with many curetted *R* genes. Interestingly, proteins from CNL-1 and TNL-4 clades did not cluster with any characterized R proteins and represented 13.5% of the predicted linseed R proteins. This indicates the unique evolutionary history and function of these proteins. The TNL-4 genes showed paraphyletic origin (Fig. 2) and their characterization might reveal some interesting and new NBS–LRR genes from linseed. All the members of CNL-2 clade clustered mostly with R proteins from various plant families, whereas CNL-3 clade clustered with mainly resistant protein RPM1 (*Arabidopsis thaliana*), indicating that these proteins are ancient in origin. The TNL-2 clade clustered with the known linseed protein P2, while the TNL-5 clade clustered with the L6 and M linseed proteins. Thus, they are the closest homologues of these characterized genes and probably represent linseed-specific NBS–LRR genes. Interestingly, TNL-3 clustered with R proteins characterized mainly from the family *Solanaceae*.

Analysis of gene expression and promoter regions

The quantitative expression analysis showed inducible expression of the RGAs. The gene model *g2659* showed the highest expression among the five gene models evaluated at 7 and 10 DAI and both control and infected tissue. This might be due to constitutive plus inducible expression of this gene. Constitutive expression of RGAs was also reported in soybean (Graham et al. 2002). The same gene also showed higher expression in control tissue, which further supports the above hypothesis. Analysis of promoter regions of the linseed NBS–LRR genes revealed uniformity in the numbers of four overrepresented *cis* elements (WBOX, CBF, DRE, and GCC boxes) across all the clades studied in both TNL and CNL families. In arabidopsis, tandemly duplicated genes show higher levels of conservation of *cis* elements compared with segmentally duplicated genes (Haberer et al. 2004). The same observation was also reported in *M. truncatula* (Ameline-Torregrosa et al. 2008). All the linseed *R* genes contained the WBOX domain and there was little difference in the frequency of occurrence of this motif between the two families (TNL and CNL). WBOX motifs are present in the *NPR1* gene (Yu and Goh 2001) and most of the pathogen response genes in arabidopsis (Li et al. 2004). It has also been shown that they activate NBS–LRR genes in arabidopsis and grape (Marchive et al. 2007; Zheng et al. 2007). The conservation of WBOX motif indicates its importance in regulation of the NBS–LRR gene family. However, it does not mean identical regulatory mechanism and other less conserved factors might be involved in fine regulation of these genes.

Based on the EST data (Table S6), 23% of the linseed NBS–LRR genes provided expression evidence, with the peeled stem library exhibiting the highest number of expressed genes. Such low expression of NBS–LRR genes has also been reported in poplar (Kohler et al. 2008), and it has been suggested that these genes are either expressed at very low levels or they are induced only during specific conditions in specific tissues. Hence, they might go undetected or escape during library preparation. Further, the expression patterns of these linseed genes could not be correlated with the presence of any of the regulatory elements or domain arrangements. However, the gene *g2659* with high EST expression contained an extra F box like domain that is known to be involved in signal transduction and protein–protein interaction. However, no correlation between gene expression and promoter sequence was observed. An in-depth analysis of such genes could reveal finer details of the disease resistance mechanism.

Conclusions

This study was undertaken to identify the NBS–LRR *R* genes using the draft genome sequence of linseed. We identified 147 NBS–LRR genes from the linseed genome having limited domain novelty, with almost all the novelty existing in the TNL subfamily, and most of that within the type D proteins. Interestingly, only a few genes were linseed specific and further characterization of these might identify novel resistance genes in linseed. Analysis of domain structure, promoter regions, phylogeny, and in silico expression revealed typical features of the NBS–LRR genes as reported from other plant species. However, some features were distinctive to linseed, including (*i*) most of the discovered linseed NBS genes are probably of ancient origin, (*ii*) only a limited variation in the domain arrangements was observed; although as reported earlier, the TNL family shows comparatively higher variation, (*iii*) uniformity of promoter regions across the gene family was detected, and (*iv*) many linseed-specific genes from the CNL and TNL families were identified. Further analysis of the complete genome sequence with accurate annotation and manual verification would provide more insight into the evolution of NBS-encoding genes in linseed. These genes can be employed to develop disease resistance varieties of linseed by molecular breeding. Additionally, this study will help to understand the disease resistance mechanisms and to generate novel disease resistance specificities in this nutritionally important crop.

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